

JOMFRULAND FYR, Norway

WMO: 014760

Lat: 58.857N

Lon: 9.576E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	14.0	17.8	7.3	8.0	18.7	11.2	9.8	23.5	30.9	38.7	27.9	38.2	9.8	30	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	74.5	63.0	71.9	62.7	69.6	61.3	66.6	71.8	64.9	69.6	63.4	68.0	8.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	90.8	69.6	62.9	86.0	68.1	61.3	81.2	66.4	31.2	72.0	29.9	69.7	28.8	68.0	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.8	27.1	24.2	13.8	79.1	4.7	3.0	10.4	81.2	7.6	82.9	4.9	84.6	1.5	86.8
			12.5	67.0	4.8	2.2	9.1	68.6	6.3	69.9	3.6	71.2	0.1	72.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.9	32.6	31.8	36.4	43.8	52.0	59.3	63.3	62.6	57.0	47.8	40.6	35.0	(6)
	DBStd	12.66	7.46	6.85	6.21	4.86	5.56	4.09	3.98	3.36	3.89	5.38	6.27	8.18	(7)
	HDD50	2565	540	511	424	194	38	0	0	0	2	107	285	464	(8)
	HDD65	6650	1005	931	888	635	405	178	82	90	241	534	732	929	(9)
	CDD50	1444	0	0	1	9	99	280	413	392	211	37	2	0	(10)
	CDD65	58	0	0	0	0	2	8	31	17	0	0	0	0	(11)
	CDH74	107	0	0	0	0	6	17	76	8	0	0	0	0	(12)
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0	(13)
	WSAvg	10.9	11.8	10.6	10.3	9.4	10.1	10.1	9.9	10.5	11.5	11.8	12.6	11.9	(14)
	PrecAvg	43.9	4.2	2.7	1.8	2.4	2.9	3.1	3.5	4.8	4.4	5.2	5.2	3.7	(15)
(16) Precipitation	PrecMax	62.5	10.5	8.8	4.8	7.0	7.9	7.2	7.7	8.7	12.2	13.7	19.3	6.7	(16)
	PrecMin	30.8	0.4	0.2	0.0	0.2	0.2	0.3	0.6	0.7	1.7	0.7	1.2	1.1	(17)
	PrecStd	8.2	2.6	1.9	1.4	1.6	1.7	1.6	1.9	2.3	2.7	2.9	3.7	1.7	(18)
	PrecStd	8.2	2.6	1.9	1.4	1.6	1.7	1.6	1.9	2.3	2.7	2.9	3.7	1.7	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	48.5	55.9	61.4	73.4	76.5	79.7	74.9	68.8	60.1	55.1	51.1	(19)
		MCWB	44.5	41.7	45.9	50.3	60.2	61.8	65.7	59.4	57.5	51.5	51.5	47.9	(20)
	2%	DB	45.7	44.4	49.6	57.0	67.8	72.2	76.0	72.6	66.0	58.3	52.9	48.9	(21)
		MCWB	42.5	39.7	42.1	47.9	58.2	61.8	65.4	61.7	57.4	52.8	50.2	46.2	(22)
	5%	DB	44.2	42.1	46.3	54.0	63.9	68.9	72.8	70.5	64.1	56.9	51.0	47.1	(23)
		MCWB	41.3	39.3	39.8	46.5	55.9	59.5	64.6	61.7	57.1	52.4	48.5	44.6	(24)
	10%	DB	42.5	40.3	43.6	51.5	60.6	66.2	70.1	68.4	62.8	55.5	49.3	45.5	(25)
		MCWB	39.7	38.0	39.3	45.2	53.4	57.7	62.6	60.8	56.6	51.2	46.9	43.1	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	43.0	46.8	52.7	63.1	66.3	69.0	67.4	62.7	57.5	52.9	48.9	(27)
		MCDB	47.2	45.3	54.6	58.1	71.0	72.6	75.4	71.7	65.5	58.3	54.2	50.3	(28)
	2%	WB	43.6	41.3	43.3	49.9	60.1	63.2	67.4	65.8	61.2	55.6	50.7	46.9	(29)
		MCDB	45.4	43.4	47.5	55.2	66.6	69.3	73.5	69.8	63.7	57.0	52.3	48.6	(30)
	5%	WB	42.0	40.0	41.6	47.9	57.3	61.2	65.8	64.6	59.9	54.1	49.0	45.3	(31)
		MCDB	43.8	41.9	44.5	52.4	62.4	67.1	71.2	68.3	62.7	56.0	50.8	46.9	(32)
	10%	WB	40.1	38.4	40.3	46.2	54.9	59.6	64.0	63.2	58.5	52.5	47.3	43.4	(33)
		MCDB	42.2	40.2	42.9	50.4	59.5	64.6	69.1	66.9	61.7	54.9	49.3	45.3	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	6.9	7.1	9.0	9.8	9.8	10.0	9.4	9.0	8.5	7.6	6.6	7.0	(35)
		MCDBR	7.7	9.4	13.5	14.4	14.7	14.1	13.4	11.0	9.9	8.4	8.0	7.3	(36)
		MCWBR	7.5	6.6	8.3	8.2	8.1	7.9	6.8	6.2	6.6	6.6	7.5	7.2	(37)
	5% WB	MCDBR	7.1	8.0	11.3	12.6	13.7	12.2	11.4	9.2	8.2	7.2	7.4	6.9	(38)
		MCWBR	7.6	6.2	7.6	7.9	8.1	7.7	6.5	6.5	6.9	7.0	7.6	7.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.286	0.320	0.340	0.379	0.372	0.357	0.378	0.377	0.349	0.336	0.302	0.263	(40)
	taud		2.240	2.321	2.369	2.325	2.415	2.478	2.438	2.438	2.506	2.444	2.299	2.175	(41)
	Ebn at Noon		169	222	254	262	273	279	269	260	251	216	161	134	(42)
	Edh at Noon		14	22	27	34	33	31	32	30	24	19	13	10	(43)
(44) All-Sky Solar Radiation	RadAvg		106	295	760	1247	1626	1858	1671	1298	842	385	141	73	(44)
	RadStd		23	62	68	176	187	136	181	86	105	66	36	9	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page